L06: Samples and Observational Studies

Statistics is Everywhere

Statistics is Everywhere

INTERNET

Online Reviews Are Biased. Here's How to Fix Them

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MARCH 06, 2018

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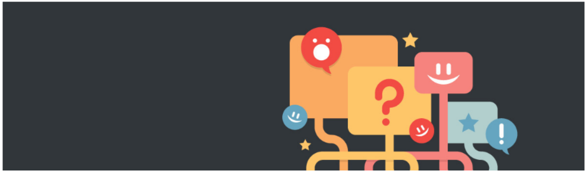
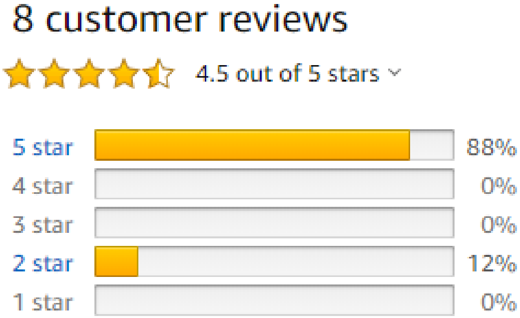


Statistics is Everywhere

- From *Harvard Business Review*, March 2018:
  Online reviews can share information at scale, but they are often written by people who voluntarily decided to post a review.
- Why might that matter for the distribution we observe?

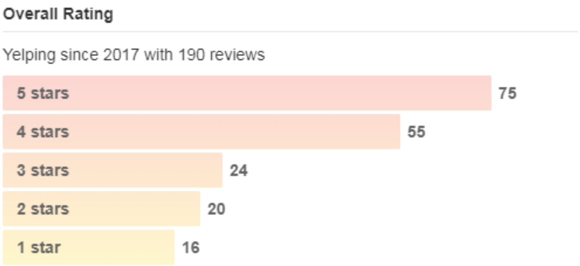
Statistics is Everywhere

- What happens to the distribution of responses when all observations are from people who volunteer to participate?
- What kind of a variable is being displayed here?
- What kind of a visualization is this?



Statistics is Everywhere

What is the sample size here?
What do we notice about this distribution?



Statistics is Everywhere

Statistics is Everywhere

- More from the article:
- Voluntary reviews often overrepresent strong opinions.
- Review distributions can become bimodal or J-shaped.
- The sample can make the product or service look more polarized than the target population of all customers.
- Sampling design affects what we think is “typical.”

Important Note

There are several definitions and descriptions in the book chapter that we will sharpen in lecture. In Edition 4, pay particular attention to:

- pg 142: Observation vs. Experiment textbox and following paragraph
- pg 143: Definition of confounding and conceptual diagram of confounding
- pg 147: Description of bias

For sampling, observational studies, study design, causality, and confounding, the material presented in lecture takes precedence.

Learning objectives for today

Defining a study by:

- Whether the treatment or exposure is controlled by an investigator
  - Experimental vs observational designs
- The population of interest
  - Target population
  - Study population
- How the sample was drawn from the population
  - Complete sample (census)
  - Random sampling
  - Convenience sampling
  - Volunteer sampling
- Was selection conditional on exposure or outcome?
- Method and timing of data collection

Who or what controls exposure?

Types of problems: from our PPDAC framework

- Recall the three types of problems we can attempt to answer:
  - Description
  - Prediction
  - Causation/Etiology
- The book argues that experiments are “the only source of fully convincing data” to study cause and effect. We disagree.
- Epidemiologists, sociologists, political scientists, economists, and others often use observational data to study cause and effect. They have developed a careful theory of causal inference that is sometimes less well understood by others.

Observation vs. Experimentation

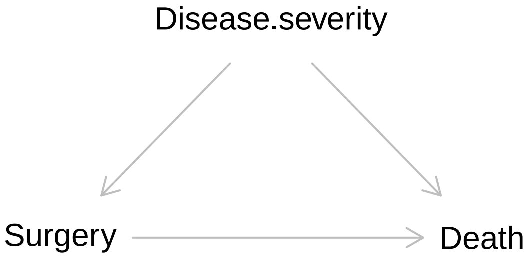
- A study is observational if we are observing what happens and do not have control over the treatments or exposures.
- A study is generally considered experimental if the investigator is experimenting by controlling who gets the exposure or treatment and who does not.

Observation vs. Experimentation

- Baldi & Moore present an example of observational data being misleading in the study of the causal effect of hormone replacement therapy on cardiovascular disease. However, these observational data, analyzed in a different way, yield the same conclusion as the randomized controlled trial. link.

Observation vs. Experimentation

- Suppose we are interested in whether a certain surgical procedure prolongs life in cancer patients:



- What is disease severity in this DAG?
- How does this DAG change if the surgical procedure is randomly assigned?

Confounders

- Remember that confounders are only important if you are asking a causal/etiologic question.
- There are no confounders for predictive studies

Other ways exposure is assigned

- Pre and post designs
- Randomized roll out (stepped wedge)
- Natural experiments

Population of interest

Target Population

- The entire group of individuals about which we want information, and to whom we would like to apply our estimates and conclusions.
- Identifying the target population is part of setting up your “problem” in our PPDAC framework

Study population

The part of the population from which we can actually select/recruit individuals and collect information.

Study sample

Individuals we were able to select/recruit and gather data from.

We use a sample to draw conclusions about the entire population.

Example: Predictors of longevity

Brandts L, van den Brandt PA. Body size, non-occupational physical activity and the chance of reaching longevity in men and women: findings from the Netherlands Cohort Study. *J Epidemiol Community Health*. Published online January 21, 2019.

- Research question: Which body size and physical activity factors are associated with reaching age 90?
- Outcome: survival to age 90
- Exposure variables: height, BMI, and non-occupational physical activity

Example: Predictors of longevity

- Data source: Netherlands Cohort Study
- Original cohort: 120,852 men and women aged 55–69 from 204 Dutch municipalities
- Analysis sample: participants born in 1916–1917 who completed a questionnaire in 1986
- Follow-up: vital status through age 90, in 2006–2007

Example: Predictors of longevity

Think about this abstract:

- What is the target population?
- What is the study population?
- To whom might we be trying to generalize these results?

Study design: Sampling

Study design: Sampling

How a sample is chosen from the population.

This is part of the “Plan” part of PPDAC

- Key design decisions:
  - Who belongs to the target population?
  - What is the sampling frame or study population?
  - Will we sample, or can we assess everyone?
  - How many individuals or units will we sample?
- Think about what you want your data frame to look like.

Complete sample: Census

- A complete sample or census includes every unit in the study population.
- A census can remove sampling variability for that study population.
- It does not automatically remove measurement error, nonresponse, or problems with the target population.

Representativeness

- Does the sample represent the population? Can we make conclusions based on the sample that will apply to the population as a whole?
- Representativeness is also called external validity.

Designs that generally do not give representative samples

- Case series
- Convenience samples
- Voluntary response samples
- Generally, non-probability designs

More representative designs: Probability samples

- Simple random sample (SRS): A sample chosen by chance, where each individual in the data set has the same chance of being selected.
- We can easily choose SRS of data frames using R

Example of SRS in R

- First read in the hospital cesarean data

```
## # A tibble: 6 x 4
##      ID Births HOSP_BEDSIZE cesarean_rate
##   <int>  <dbl>        <dbl>         <dbl>
## 1     1    767            1         0.344
## 2     2    183            1         0.454
## 3     3    668            1         0.430
## 4     4    154            1         0.279
## 5     5    327            1         0.306
## 6     6   2356            1         0.301
```

Example of SRS in R

```
CS_100_1 <- CS_data %>% slice_sample(n = 100)
CS_100_2 <- CS_data %>% slice_sample(n = 100)
```

Do you expect `head(CS_100_1)` to equal `head(CS_100_2)`?

Example of SRS in R

```
head(CS_100_1 %>% select(Births, HOSP_BEDSIZE, cesarean_rate, ID))
```

```
## # A tibble: 6 x 4
##   Births HOSP_BEDSIZE cesarean_rate    ID
##    <dbl>        <dbl>         <dbl> <int>
## 1   2839            2         0.333   154
## 2    185            2         0.357   178
## 3   1246            3         0.220   576
## 4    767            1         0.344     1
## 5    412            2         0.221   148
## 6    273            2         0.366   307
```

Example of SRS in R

```
head(CS_100_2 %>% select(Births, HOSP_BEDSIZE, cesarean_rate, ID))
```

```
## # A tibble: 6 x 4
##   Births HOSP_BEDSIZE cesarean_rate    ID
##    <dbl>        <dbl>         <dbl> <int>
## 1   3598            3         0.360   433
## 2    442            3         0.308   358
## 3   1103            2         0.358   196
## 4    516            3         0.298   474
## 5   3683            3         0.193   346
## 6   2945            3         0.399   560
```

Example of SRS in R

```r
identical(CS_100_1, CS_100_2)
```

```
## [1] FALSE
```

Example of SRS in R

Why are these first six lines different?

Anytime you do something *randomly* in R, the results will be different. This is a good thing! This allows you to pick many different random samples. In future weeks we will do this a lot.

Example of SRS in R

What if you want to ensure that you pick the same SRS as a friend?

Then you need to use `set.seed()`:

```
set.seed(123)
CS_100_1 <- CS_data %>% slice_sample(n = 100)

set.seed(123)
CS_100_2 <- CS_data %>% slice_sample(n = 100)

identical(CS_100_1, CS_100_2)
```

```
## [1] TRUE
```

SRS: a fraction in R

```r
CS_5percent <- CS_data %>% slice_sample(prop = 0.05)
```

Proportionate stratified sampling in R

- Group *and then* sample the same fraction from each group

```
CS_10percent_grouped <- CS_data %>%
  group_by(HOSP_BEDSIZE) %>%
  slice_sample(prop = 0.1)
dim(CS_10percent_grouped)
```

```
## [1] 57  8
```

- Proportionate stratified SRS assembles a sample that maintains the relative proportions of `HOSP_BEDSIZE` in the chosen sample compared to the population

Proportionate stratified sampling in R

How to check you really did sample 10% of each `HOSP_BEDSIZE` group?

First see how many hospitals fall into each category in the original data

```
CS_data %>%
  count(HOSP_BEDSIZE)
```

```
## # A tibble: 3 x 2
##   HOSP_BEDSIZE     n
##          <dbl> <int>
## 1            1   131
## 2            2   179
## 3            3   270
```

Proportionate stratified sampling in R

Then in the sample:

```
CS_10percent_grouped %>%
  ungroup() %>%
  count(HOSP_BEDSIZE)
```

```
## # A tibble: 3 x 2
##   HOSP_BEDSIZE     n
##          <dbl> <int>
## 1            1    13
## 2            2    17
## 3            3    27
```

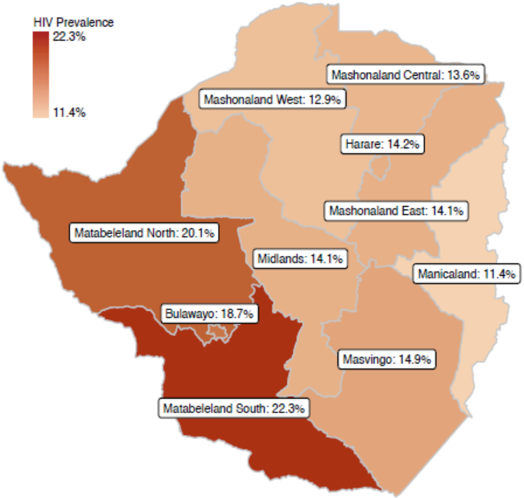
Disproportionate stratified sampling in R

- When might you want to overrepresent certain groups?
- Example: Estimating infant mortality by race/ethnicity when some race/ethnic groups are very small (e.g., indigenous groups in U.S./Canada)
- Then, you may want to oversample certain groups so you can better estimate infant mortality in those groups than if you sampled proportionately

Multistage sampling

- Random samples inside random samples
- For example, sampling schools using an SRS, and then sampling students within those schools.

Multistage sampling



Conditional selection?

- Was selection conditional on exposure or outcome?
- Selection based on exposure status:
  - Usually a cohort design
- Selection based on outcome status:
  - Usually a case-control design
- Selection based on multiple factors:
  - Matched-pair or matched-set designs

Conditional selection?

Why it matters:

- If selection is conditional on exposure, the marginal distribution of exposure is not meaningful.
- If selection is conditional on outcome, the marginal distribution of outcome is not meaningful.
- The analysis must match the way the sample was selected.

Study design: Data collection

Study design: Data collection

- Part of your study design is also about how you will collect the variables:
  - By survey? (self-reported information)
  - Measured by device?
  - Collected from health records?
  - Google search?
- What are the types of variables and levels of each variable?
  - Continuous?
  - Categorical? How many categories? Ordinal or nominal?
- Think about what you want your data frame to look like.

Temporality

Temporality

- Method and timing of data collection
  - One timepoint only (cross sectional)
  - Multiple timepoints (longitudinal)
  - Moving forward in time (prospective)
  - Collecting data from previous times (retrospective)

Putting it together

Reading an article title:
What does the title tell you about the design? What would you ask about the methods?

From the *American Journal of Epidemiology*, Volume 188, Issue 2, February 2019:

- Associations of a Healthy Lifestyle Index With the Risks of Endometrial and Ovarian Cancer Among Women in the Women's Health Initiative Study
- Death and Chronic Disease Risk Associated With Poor Life Satisfaction: A Population-Based Cohort Study

Putting it together

From *New England Journal of Medicine* 2019; 380:415-424:

- Partial Oral versus Intravenous Antibiotic Treatment of Endocarditis: a Randomized, Noninferiority, Multicenter Trial

Ask:

- What is the exposure or treatment?
- Who is the target population?
- How were participants selected and followed?

R code recap

We used two functions to sample from a dataset:

- `slice_sample(n = ...)`
- `slice_sample(prop = ...)`

Next lecture

- More about trials
- Sources of Bias

Parting humor

From PhDcomics.com